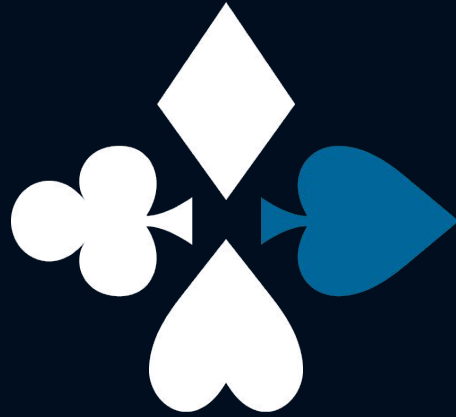




Pokerbots 2026

Lecture 4: Machine Learning

Sponsors



CITADEL | CITADEL Securities



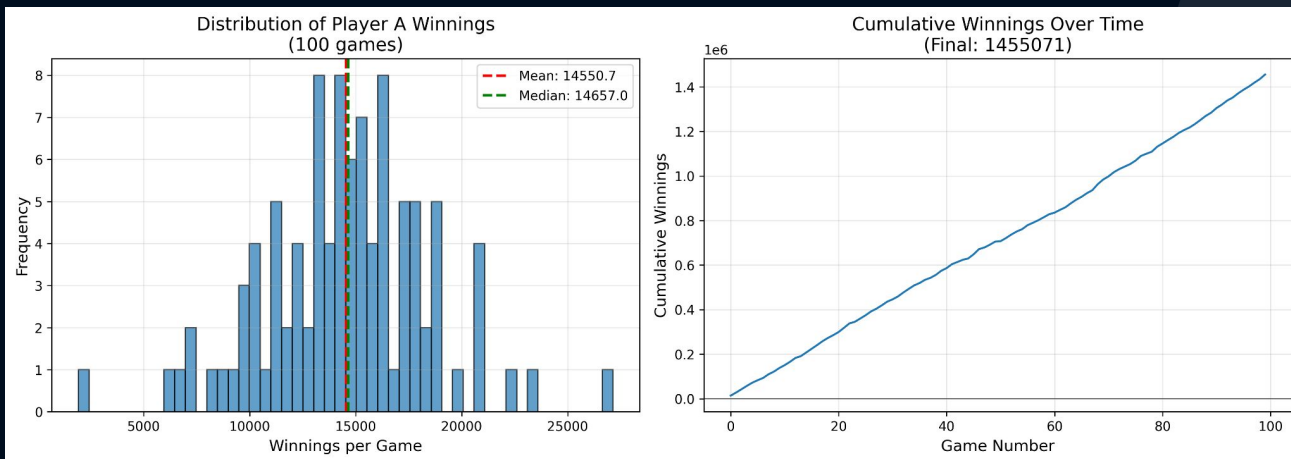


Announcements

Reference Bot 3 Posted

- See pkr.bot/refbot3
- Tosses card based on a Monte Carlo simulation
- Bets high when hole cards are high

Note: At least against the skeleton bot, this bot performs slightly worse than refbot2 (but very similarly)! Why might this be?



Hackathon Next Week!

- Wednesday night (1/14)
- Show up and grind
- Dinner provided
- Snacks, fun, and games
- Prizes for challenges and those that stick around
- Details will be announced on Piazza

No Class Next Thursday

Cancelled in light of Hackathon the night before

The background is a solid blue color. On the left side, there are several overlapping circles of varying shades of blue, creating a decorative, abstract pattern.

Giveaway Time!

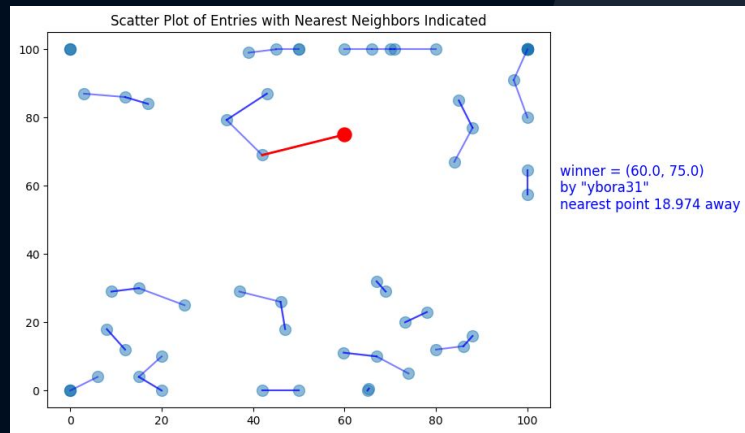
Distance Game: pk_r.bot/dist

- Pick a point (x, y) , x and y are reals from 0-100
- Winner is whoever's the furthest from anyone else's guess (maximum distance to nearest neighbor)

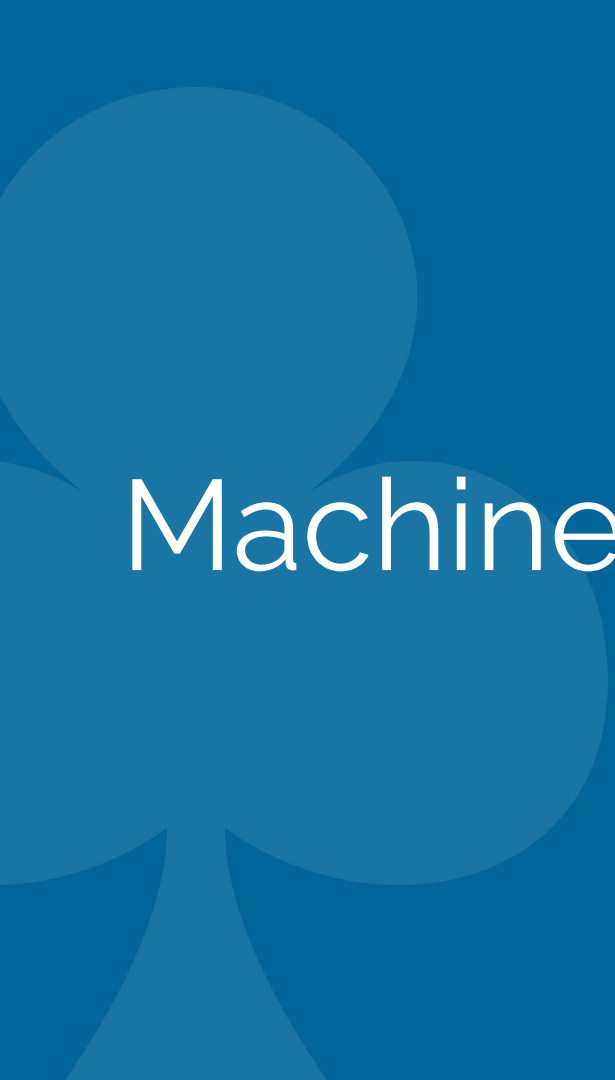


BEATS STUDIO PRO

(if winner doesn't claim, we go down list in order of distance)



from last year^

The background is a solid blue color. On the left side, there are several overlapping circles of varying shades of blue, creating a decorative pattern.

Machine Learning

Agenda for Today

- Machine Learning High Level Overview
- Problem Setting for Supervised Learning
- Is This Even Statistics
- Foundational Algorithms
- Creating a Good Model



High Level Overview

What is Machine Learning (ML)?

- Using a machine (computation) to find patterns in data
 - Pattern = ???
- Found patterns can be applied without explicit programming
- Key idea: *generalize* instead of *memorize*
- Typically utilize large amounts of data
- Used for prediction, data compression, automation, decision-making

Typical ML Workflow

- Acquire dataset
- Create model(s)
- Evaluate model(s)
- Hopefully have good model via selection and improvement
- Deploy!

Types of Learning

- Supervised Learning
 - Dataset has examples of correct behavior
 - “Questions with answers”
 - Prediction tasks
- Unsupervised Learning
 - Dataset has no notion of good behavior
 - Task is to find patterns to represent dataset
 - Clustering, compression, generative modeling
- Reinforcement Learning (RL)
 - Environment with feedback (rewards), but no correct target
 - Sequential decision making, control, pokerbots!

Questions?



Problem Setting

Example

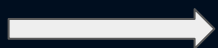
- Supervised learning setting (which we will focus on today)
- AI detection software is really big nowadays
 - Maybe we can make an awesome one and get super rich >:D
- So let's make our algorithm take a photo and tell us if it's a human or a robot
- We have example photos and labels of both

Example

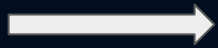
Data



“Human”



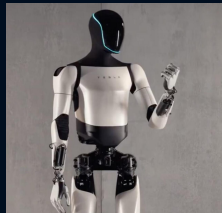
“Human”



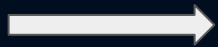
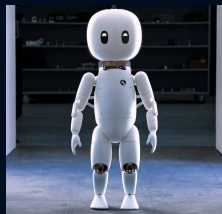
“Human”

Labels

Example



“Robot”



“Robot”



“Robot”

Example

It's a photorealistic humanoid robot (or android) designed to look like a human, likely for robotics or AI demonstration purposes.

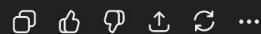
📄 👍 🗣️ ⬆️ ↻ ...



Human or Robot?

Example

It's a person outdoors mid-movement (likely stretching or exercising) with their face obscured by a robot emoji.



Human or Robot?

Key Variables in Supervised Learning

- data example contains
 - “input” properties called *features*
 - “output” property called *label*
- *dataset* is a collection of these examples
- an *algorithm* that takes as an input an incomplete example with only features, and outputs a label to complete the example.

Types of Prediction

Classification

- Predict a category
- Example: “human or robot”
- **Discrete labels**

Regression:

- Predict value aka real number(s)
- Example: giveaway estimations
- **Continuous labels**

Questions?



Is This Even Statistics

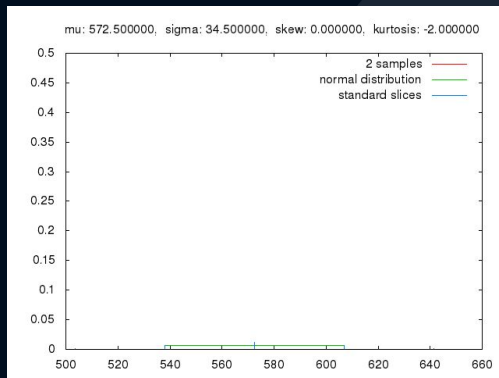
Stats Refresher

- Given some data, try to infer information about the distribution
 - Sounds vaguely similar to this ML stuff
- The bridge:
 - Given past data, predict future data
 - Even more similar to ML
- Probability + Statistics = ML?

Not exactly

Motivating Supervised Learning ($X \rightarrow Y$)

- Suppose we have no inputs, just want to predict output
 - Output will follow some distribution
 - Sample distribution $\rightarrow$ Population distribution (Glivenko-Cantelli)
 - $P(Y)$ (here, Y is a random variable)
 - For predicting future, maybe use EV?
- Recall Subjectivist/Bayesian interpretation of statistics
- Now, with inputs = x , we can:
 - Start with prior output distribution
 - Apply Bayesian update from the inputs
 - $P(Y | X = x)$ (here, X and Y are random variables, x is the current value of the inputs)
- Essentially, we want to estimate $P(Y | X)$, or the distribution of outputs given some inputs!



Motivating Supervised Learning ($X \rightarrow Y$)

- Supervised Learning = Bayesian inference over predictive mappings
 - labels act as observations that update a prior belief about how inputs generate outputs
- The core task
 - statistical inference about the **structure of the data-generating process itself**
 - NOT just the relationship between inputs and outputs

Briefly Expanding to ML in General

- Unsupervised Learning

- about inferring the hidden structure or generative process of the data using **only the data itself**
- guided by **inductive biases** encoded in the model
- begin with a prior over latent structure that could have produced the data
- update it using observations alone

- Reinforcement Learning

- inference under delayed/partial feedback
- rewards constrain which world models and action strategies are plausible
- NOT directly specifying correct outputs

- OVERALL

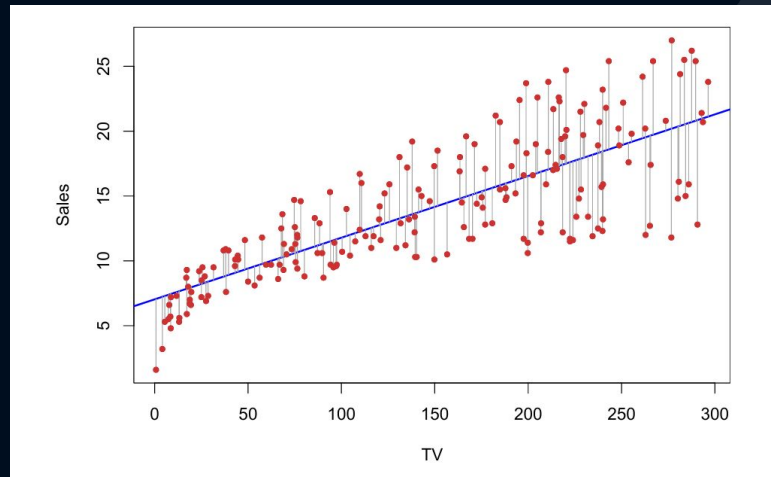
- All machine learning is about *inferring structure in the world from data*, guided by the form of **feedback** we are given - labels, rewards, or the data itself.



Foundational Algorithms

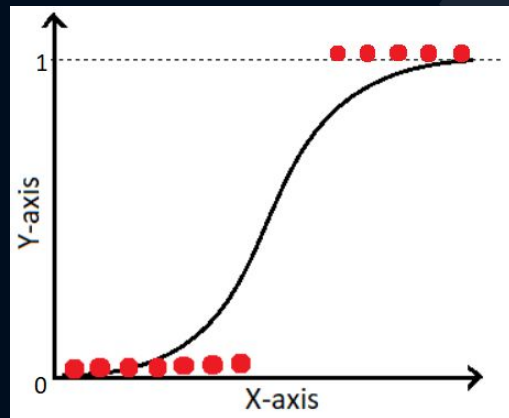
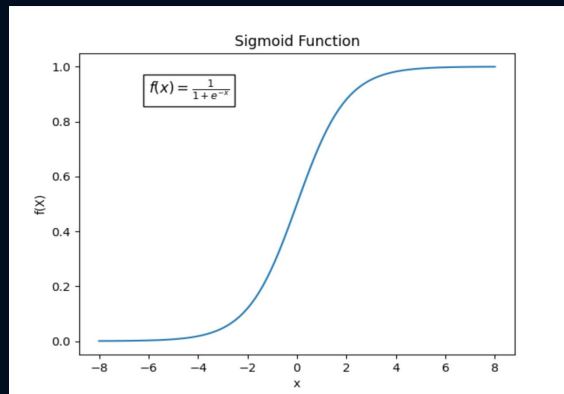
Linear Regression

- Linear relation between input and output
- “Line of best fit”
- $y = w \cdot x + b$
- Can generalize to multiple variables
 - $y = w_1 \cdot x_1 + w_2 \cdot x_2 + \dots + b$
- Easy to work with but only works well under certain conditions



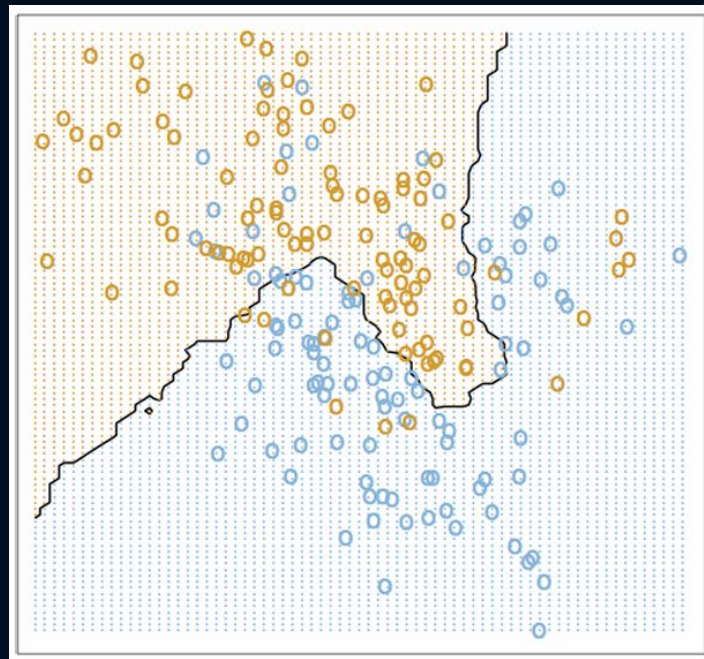
Logistic Regression

- Modifies linear regression for binary classification
- Sigmoid function $\sigma(x)$ transforms any real number onto interval (0,1).
- $y = \sigma(w \cdot x + b)$
- Interpreted as outputting a probability of belonging to a specified class



k-Nearest Neighbors (kNN)

- Classification or regression
- Decision using proximity
- For an input:
 - Find the k closest neighbors within dataset
 - Aggregate their labels to decide output
 - Majority vote for classification
 - Average for regression
- Very flexible and can represent complex patterns
- Struggles with higher dimensions



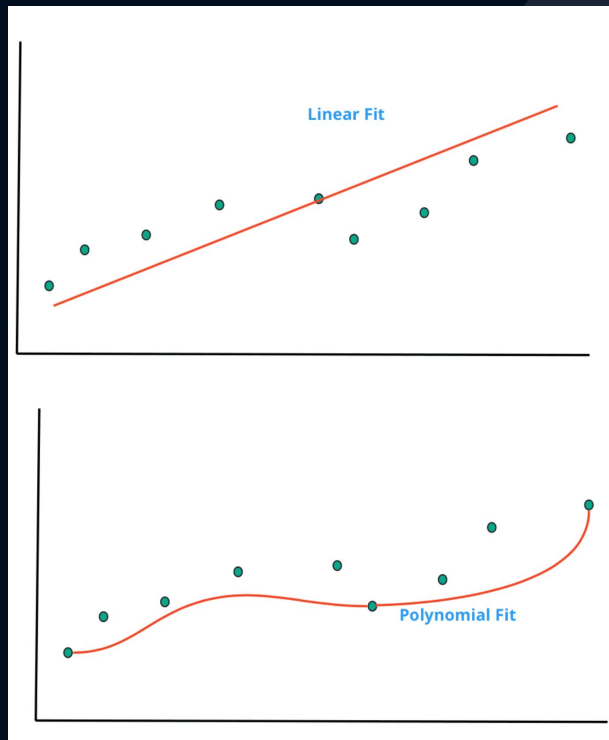
Questions?

Creating a Good Model

Creating a Good Model

What is a model?

- Template for an algorithm
- *Model class* or *hypothesis class* is the collection of all possible algorithms that follow that template
- Some models are defined by *parameters*
 - Ex: w and b in linear regression
 - Space of possible parameters $\leftrightarrow$ model class
 - Choosing a model equates to choosing the parameters
- Other models are *non-parametric* (kNN)
- The larger the model class, the more *expressive* a model is



Creating a Good Model

Loss Functions

- Metric for seeing if model is good by measuring the error it makes on an example
- This equates to measuring some *distance* between predicted and true label
 - Linear Regression: squared distance
 - Logistic Regression: cross-entropy
 - kNN: 0/1 accuracy rate
- Loss function reports a penalty for a single input-output example - model evaluates based on cumulative or average loss across whole set of examples

How Important is Choice of Loss Function?

- Do different functions (i.e. dist vs. dist^2) cause different model behavior?
- Recall that for input = x , we want to predict information about $P(Y | X = x)$
 - Generally, we make our models output deterministic values (i.e. a statistic about $P(Y | X = x)$)
 - To get the lowest dist^2 assuming that our distribution is right, we would output the mean
 - To get the lowest dist assuming that our distribution is right, we would output the median
- Even though two loss functions might have similar behavior (dist and dist^2 rank deviations similarly), they can cause different model behavior!

Creating a Good Model

Lunch 🤪

Leave any type of feedback at [pkr.bot/feedback!](https://pkr.bot/feedback)



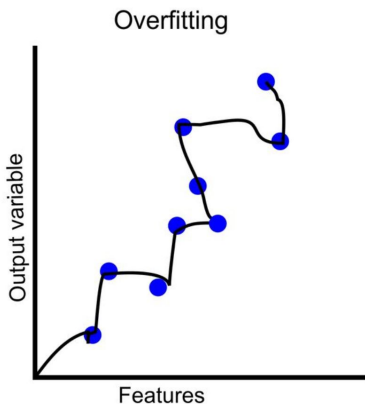
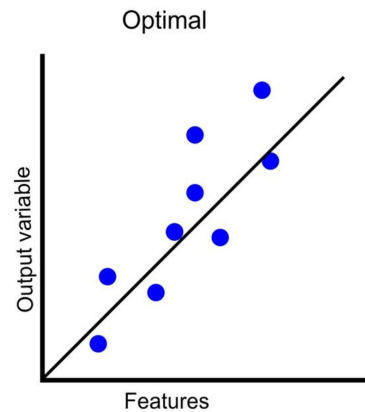
Creating a Good Model

Combine everything together!

- Consider all models within the hypothesis class and pick the one that has the best (lowest) loss across the entire dataset
- For parametric models, this is a minimization problem over the parameter space
 - “Training”
 - Sometimes near impossible to find the absolute best one so approximations are settled for
- Why don't we simply just use the most expressive hypothesis class? More options means that the best options are going to have better results...
 - Ex: the space of polynomial functions can do more than everything linear functions can

Overfitting

- “Too good to be true”
- Model fits dataset extremely well but fails to capture underlying relationship (which is what we want)
- Memorization occurs, which prevents generalization
- Caused by overexpressive hypothesis class, aka too many parameters, relative to number of data points



Inductive Bias

- the set of assumptions or preferences a model uses to **generalize**
- A whole different beast
 - requires analyzing the nature of the data + finding assumptions that are **not too limiting**
 - i.e. linear regression is great sometimes, but not always
 - hard since tough problems are often abstract and crafting assumptions is difficult
 - *architecture research*
- Modern ML has shifted from explicitly designing inductive biases to relying on scale to implicitly learn them
 - works in-distribution but often fails when data or objectives change
 - lowkey pretty sad to see in some cases

Model Evaluation

- Split dataset into training and testing groups
 - Create model only using train set
 - Evaluate model based on loss across test set
- Goal is to detect overfitting and have a more realistic measure of a good model
- Perform the split randomly
 - Each data point has equal chance of showing up in either
 - Idea is to have the data be from the same distribution but still different
- Analogy: Practice exams vs Real exam
 - Test same capabilities using different questions
 - If real exam was exact copy of practice, then doing well doesn't say much about what was learned
- When we deploy model it may fail - we want to assess whether this would happen (ideally not)

Sources of Error

Bias

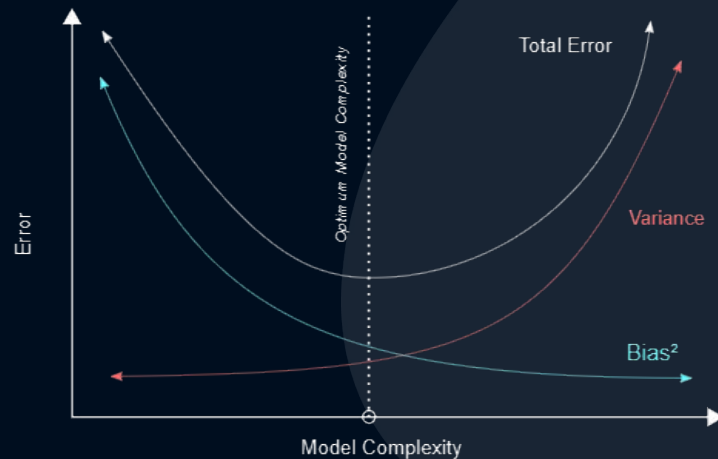
- error due to simplistic assumptions
- model class not expressive/flexible enough to represent data
- consistent error resulting from approach
- Characterized by high loss in both training and testing
- Ex: Linear Regression

Variance

- error from sensitivity to training data
- Model class is overly flexible, attempting to capture noise as a pattern
- Characterized by low loss in training but high loss in testing (aka overfitting)
- Ex: kNN

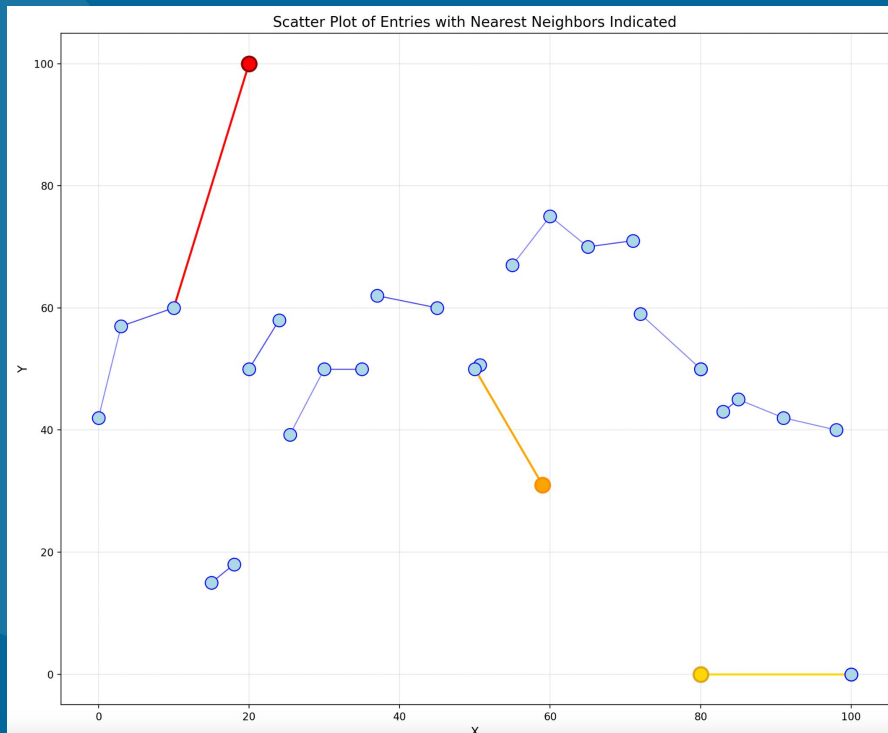
Bias-Variance Tradeoff

- There is an inherent tradeoff between bias and variance within model selection
- Many design decisions (aka hyperparameters) increase/decrease the model's complexity which could lead to too much bias or variance
- Some considerations to find the “sweet spot”:
 - Validation sets
 - (Fail fast)
 - Regularization
 - (A type of induction bias)
 - Careful feature selection/pruning
 - (A type of induction bias)



Questions?

Distance Game: kerbs!



1st: chriskc
2nd: hnagai
3rd: zeng02



Thanks for watching!

Information hub at pkr.bot/hub

Slides/notes will be posted on pkr.bot/resources

Make sure to check pkr.bot/piazza for updates

Lecture recordings at pkr.bot/panopto

Leave feedback at pkr.bot/feedback!